

Functional & Performance Testing Template
Model Performance Test

Date	19 July 2026
Team ID	
Project Name	AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Loan Input Validation (outstanding amount, EMI, overdue months, income)	Enter valid numeric values, then try negative numbers, blanks, and non-numeric text in the Add Loan form	Valid values are accepted and the loan is created; invalid values are rejected with a clear error before submission		
FT-02	Debt-Stress Score & Settlement % Calculation	Create loans with varying EMI ratios and overdue months; open the loan's analysis view	EMI ratio, monthly surplus, debt-stress level (Low/Moderate/High/Critical) and settlement % match the documented formula and clamp range (35-75%)		
FT-03	Negotiation Letter Generation	Select a letter type and tone on a loan, click "Generate," and review the output	A personalized letter is generated referencing the borrower's actual loan numbers and chosen tone		
FT-04	Gemini API Connection Check	Generate a letter with a valid Gemini API key configured, then again with the key removed/invalid	With a valid key, Gemini generates the letter; without one, the system falls back to the template generator without crashing		
FT-05	Dashboard Aggregation	Add multiple loans with different stress levels, then open the dashboard	Dashboard summary and settlement history correctly aggregate all loans		
PT-01	Response Time - Loan Analysis	Time how long /loans/{id}/analysis takes to return after a request	Should return in under 1 second (pure rule-based calculation, no external call)		
PT-02	Response Time - Letter Generation	Time how long /negotiation/generate takes to return a Gemini-generated letter	Should complete in under 5-8 seconds under normal Gemini API latency; template fallback should return in under 1 second		
PT-03	Concurrent Request Load Test	Send multiple simultaneous requests to /loans and /negotiation/generate	API remains responsive and does not error out or significantly slow down under concurrent load		
PT-04	Dashboard Load with Multiple Loans	Populate the database with a large number of loans (e.g. 100+) and load the dashboard	Dashboard summary loads smoothly without noticeable lag		

Test Steps and Expected Result columns are filled in based on the project's documented behavior. Actual Result and Pass/Fail should be completed after running each test against the deployed application.